CLIMATE CHANGE SCIENCE GREENHOUSE EFFECT

GREENHOUSE-EFFECT FIREWALL → ENERGY-BALANCE RAIL → EMISSION-TO-CONCENTRATION LEDGER → FORCING-FEEDBACK FORK → WATER AND PARTICLE MATRIX → GAS COMPARISON GATE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 GREENHOUSE-EFFECT FIREWALL

GREENHOUSE-EFFECT FIREWALL | NATURAL EFFECT | ABSORBS AND RE-EMITS OUTGOING LONGWAVE RADIATION | MAINTAINS A HABITABLE ENERGY BALANCE | HUMAN ENHANCEMENT | RAISES CONCENTRATIONS AND NET FORCING | CLIMATE RESPONSE | WARMING PLUS WIDER SYSTEM CHANGES

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
NATURAL EFFECT → absorbs and re-emits outgoing longwave radiation	HUMAN ENHANCEMENT → raises concentrations and net forcing	EXAM RULE → preserve the natural-versus-enhanced distinction
FUNCTION → maintains a habitable energy balance	CLIMATE RESPONSE → warming plus wider system changes	NATURAL EFFECT → absorbs and re-emits outgoing longwave radiation

ECOLOGICAL UNIT / PARAMETER
- NATURAL EFFECT → absorbs and re-emits outgoing longwave radiation
- FUNCTION → maintains a habitable energy balance

DECISIVE DISTINCTION
- HUMAN ENHANCEMENT → raises concentrations and net forcing
- CLIMATE RESPONSE → warming plus wider system changes

SCALE / STATUS / EXCEPTION
- EXAM RULE → preserve the natural-versus-enhanced distinction
- NATURAL EFFECT → absorbs and re-emits outgoing longwave radiation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLIMATE RESPONSE → warming plus wider system changes.

01

STAGE 01 ENERGY-BALANCE RAIL

ENERGY-BALANCE RAIL | INCOMING PREDOMINANTLY SHORTWAVE ENERGY | ABSORBS PART AND WARMS | EMITS OUTGOING LONGWAVE INFRARED ENERGY | GREENHOUSE GASES | ABSORB AND RE-EMIT PART OF LONGWAVE FLOW | FORCING CHANGES ENERGY RETAINED BY THE CLIMATE SYSTEM

SUN → incoming predominantly shortwave energy → SURFACE → absorbs part and warms → EARTH → emits outgoing longwave infrared energy → GREENHOUSE GASES → absorb and re-emit part of longwave flow → IMBALANCE → forcing changes energy retained by the climate system

SYSTEM INPUT / STARTING CONDITION
- SUN → incoming predominantly shortwave energy
- SURFACE → absorbs part and warms

ECOLOGICAL PROCESS / TRANSFER
- EARTH → emits outgoing longwave infrared energy
- GREENHOUSE GASES → absorb and re-emit part of longwave flow

OUTCOME / FEEDBACK / LIMIT
- IMBALANCE → forcing changes energy retained by the climate system
- SUN → incoming predominantly shortwave energy

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: IMBALANCE → forcing changes energy retained by the climate system.

02

STAGE 02 EMISSION-TO-CONCENTRATION LEDGER

EMISSION-TO-CONCENTRATION LEDGER | EMISSIONS FLOW DURING A STATED PERIOD | CONCENTRATION AFTER SOURCES AND SINKS INTERACT | LONG-LIVED GASES BUILD AN ATMOSPHERIC STOCK | FLOW CUT | SLOWS ADDITION BUT DOES NOT ERASE THE STOCK | CHANGES THE STOCK ONLY WITHIN A STATED BOUNDARY

ACTIVITY → emissions flow during a stated period

ATMOSPHERE → concentration after sources and sinks interact → ACCUMULATION → long-lived gases build an atmospheric stock → FLOW CUT → slows addition but does not erase the stock

SYSTEM / LEVEL / DEFINITION
- ACTIVITY → emissions flow during a stated period
- ATMOSPHERE → concentration after sources and sinks interact

STRUCTURE-FUNCTION MECHANISM
- ACCUMULATION → long-lived gases build an atmospheric stock
- FLOW CUT → slows addition but does not erase the stock

SCALE / EVIDENCE LIMIT
- REMOVAL → changes the stock only within a stated boundary
- ACTIVITY → emissions flow during a stated period

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REMOVAL → changes the stock only within a stated boundary.

03

STAGE 03 FORCING-FEEDBACK FORK

FORCING-FEEDBACK FORK | EXTERNAL OR IMPOSED ENERGY CHANGE | RADIATIVE FORCING | CLIMATE RESPONSE TRIGGERED | AMPLIFIES INITIAL CHANGE | POSITIVE FEEDBACK | DAMPENS INITIAL CHANGE | NEGATIVE FEEDBACK

EXTERNAL OR IMPOSED ENERGY CHANGE → radiative forcing

CLIMATE RESPONSE TRIGGERED → feedback → AMPLIFIES INITIAL CHANGE → positive feedback → DAMPENS INITIAL CHANGE → negative feedback

SYSTEM / LEVEL / DEFINITION
- ACTIVITY → emissions flow during a stated period
- ATMOSPHERE → concentration after sources and sinks interact

STRUCTURE-FUNCTION MECHANISM
- ACCUMULATION → long-lived gases build an atmospheric stock
- FLOW CUT → slows addition but does not erase the stock

SCALE / EVIDENCE LIMIT
- REMOVAL → changes the stock only within a stated boundary
- ACTIVITY → emissions flow during a stated period

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REMOVAL → changes the stock only within a stated boundary.

03

STAGE 03 FORCING-FEEDBACK FORK

FORCING-FEEDBACK FORK · EXTERNAL OR IMPOSED ENERGY CHANGE · RADIATIVE FORCING · CLIMATE RESPONSE TRIGGERED · AMPLIFIES INITIAL CHANGE · POSITIVE FEEDBACK · DAMPENS INITIAL CHANGE · NEGATIVE FEEDBACK

EXTERNAL OR IMPOSED ENERGY CHANGE → radiative forcing

CLIMATE RESPONSE TRIGGERED → feedback → AMPLIFIES INITIAL CHANGE → positive feedback → DAMPENS INITIAL CHANGE → negative feedback

SYSTEM INPUT / STARTING CONDITION
- EXTERNAL OR IMPOSED ENERGY CHANGE → radiative forcing
- CLIMATE RESPONSE TRIGGERED → feedback

ECOLOGICAL PROCESS / TRANSFER
- AMPLIFIES INITIAL CHANGE → positive feedback
- DAMPENS INITIAL CHANGE → negative feedback

OUTCOME / FEEDBACK / LIMIT
- SIGN → direction of response, never moral value
- EXTERNAL OR IMPOSED ENERGY CHANGE → radiative forcing

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SIGN → direction of response, never moral value.

04

STAGE 04 WATER AND PARTICLE MATRIX

WATER AND PARTICLE MATRIX · WATER VAPOUR · GREENHOUSE GAS COMMONLY ACTING AS FEEDBACK · SULPHATE-LIKE AEROSOLS · OFTEN COOLING INFLUENCE · BLACK CARBON · ABSORBING WARMING INFLUENCE · SNOW OR ICE DEPOSITION

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
WATER VAPOUR → greenhouse gas commonly acting as feedback	BLACK CARBON → absorbing warming influence	BOUNDARY → agent, location and mechanism determine the sign
SULPHATE-LIKE AEROSOLS → often cooling influence	SNOW OR ICE DEPOSITION → can lower albedo	WATER VAPOUR → greenhouse gas commonly acting as feedback

ECOLOGICAL UNIT / PARAMETER
- WATER VAPOUR → greenhouse gas commonly acting as feedback
- SULPHATE-LIKE AEROSOLS → often cooling influence

DECISIVE DISTINCTION
- BLACK CARBON → absorbing warming influence
- SNOW OR ICE DEPOSITION → can lower albedo

SCALE / STATUS / EXCEPTION
- BOUNDARY → agent, location and mechanism determine the sign
- WATER VAPOUR → greenhouse gas commonly acting as feedback

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BOUNDARY → agent, location and mechanism determine the sign.

05

STAGE 05 GAS COMPARISON GATE

GAS COMPARISON GATE · RELATIVE INTEGRATED INFLUENCE OVER A NAMED HORIZON · EFFECT PER UNIT MASS UNDER THE METRIC · PERSISTENCE IN THE ATMOSPHERE · EMISSION SCALE · QUANTITY RELEASED · NEAR-TERM AND CUMULATIVE LEVERS ARE NOT IDENTICAL

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
GWP → relative integrated influence over a named horizon	LIFETIME → persistence in the atmosphere	POLICY → near-term and cumulative levers are not identical
POTENCY → effect per unit mass under the metric	EMISSION SCALE → quantity released	GWP → relative integrated influence over a named horizon

ECOLOGICAL UNIT / PARAMETER
- GWP → relative integrated influence over a named horizon
- POTENCY → effect per unit mass under the metric

DECISIVE DISTINCTION
- LIFETIME → persistence in the atmosphere
- EMISSION SCALE → quantity released

SCALE / STATUS / EXCEPTION
- POLICY → near-term and cumulative levers are not identical
- GWP → relative integrated influence over a named horizon

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: POLICY → near-term and cumulative levers are not identical.

06

STAGE 06 WEATHER-TO-CLIMATE LADDER

WEATHER-TO-CLIMATE LADDER · INDIVIDUAL WEATHER OCCURRENCE · SEASON OR YEAR · SHORT-PERIOD VARIABILITY · FREQUENCY, INTENSITY AND PATTERN · LONGER RECORD · CLIMATE TREND ASSESSMENT · FLUCTUATION AROUND A TREND IS NOT TREND REVERSAL

EVENT → individual weather occurrence

SEASON OR YEAR → short-period variability → DISTRIBUTION → frequency, intensity and pattern → LONGER RECORD → climate trend assessment

ECOLOGICAL UNIT / PARAMETER
- GWP → relative integrated influence over a named horizon
- POTENCY → effect per unit mass under the metric

DECISIVE DISTINCTION
- LIFETIME → persistence in the atmosphere
- EMISSION SCALE → quantity released

SCALE / STATUS / EXCEPTION
- POLICY → near-term and cumulative levers are not identical
- GWP → relative integrated influence over a named horizon

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: POLICY → near-term and cumulative levers are not identical.

06

STAGE 06 WEATHER-TO-CLIMATE LADDER

WEATHER-TO-CLIMATE LADDER · INDIVIDUAL WEATHER OCCURRENCE · SEASON OR YEAR · SHORT-PERIOD VARIABILITY · FREQUENCY, INTENSITY AND PATTERN · LONGER RECORD · CLIMATE TREND ASSESSMENT · FLUCTUATION AROUND A TREND IS NOT TREND REVERSAL

EVENT → individual weather occurrence

SEASON OR YEAR → short-period variability → DISTRIBUTION → frequency, intensity and pattern → LONGER RECORD → climate trend assessment

SYSTEM / LEVEL / DEFINITION
- EVENT → individual weather occurrence
- SEASON OR YEAR → short-period variability

STRUCTURE-FUNCTION MECHANISM
- DISTRIBUTION → frequency, intensity and pattern
- LONGER RECORD → climate trend assessment

SCALE / EVIDENCE LIMIT
- RULE → fluctuation around a trend is not trend reversal
- EVENT → individual weather occurrence

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → fluctuation around a trend is not trend reversal.

07

STAGE 07 ATTRIBUTION EVIDENCE CHAIN

ATTRIBUTION EVIDENCE CHAIN · DEFINE THE MEASURED CHANGE · DISTINGUISH SIGNAL FROM EXPECTED VARIABILITY · COMPARE HUMAN AND NATURAL INFLUENCES · MODELS PLUS PHYSICS PLUS RECORDS · CONVERGING EVIDENCE · QUALIFIED CONTRIBUTION, NOT SLOGAN CAUSATION

OBSERVATION → define the measured change → DETECTION → distinguish signal from expected variability → DRIVERS → compare human and natural influences → MODELS PLUS PHYSICS PLUS RECORDS → converging evidence → ATTRIBUTION → qualified contribution, not slogan causation

SYSTEM INPUT / STARTING CONDITION
- OBSERVATION → define the measured change
- DETECTION → distinguish signal from expected variability

ECOLOGICAL PROCESS / TRANSFER
- DRIVERS → compare human and natural influences
- MODELS PLUS PHYSICS PLUS RECORDS → converging evidence

OUTCOME / FEEDBACK / LIMIT
- ATTRIBUTION → qualified contribution, not slogan causation
- OBSERVATION → define the measured change

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MODELS PLUS PHYSICS PLUS RECORDS → converging evidence.

08

STAGE 08 EVENT-CLAIM FIREWALL

EVENT-CLAIM FIREWALL · DEFINE EVENT CLASS · PLACE, DURATION AND METRIC · COMPARE WORLDS · WITH AND WITHOUT HUMAN INFLUENCE · ESTIMATE CHANGE · PROBABILITY OR INTENSITY · RETAIN OTHER DRIVERS

SYSTEM / LEVEL / DEFINITION
- DEFINE EVENT CLASS → place, duration and metric
- COMPARE WORLDS → with and without human influence

STRUCTURE-FUNCTION MECHANISM
- ESTIMATE CHANGE → probability or intensity
- RETAIN OTHER DRIVERS → exposure, vulnerability and variability

SCALE / EVIDENCE LIMIT
- NO SOLE CAUSE → disaster impact is multi-causal
- DEFINE EVENT CLASS → place, duration and metric

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NO SOLE CAUSE → disaster impact is multi-causal.

09

STAGE 09 OBSERVATION-PROJECTION MATRIX

OBSERVATION-PROJECTION MATRIX · MEASURED PAST OR PRESENT · EXPLANATION OF OBSERVED CHANGE · CONDITIONAL ASSUMPTIONS · MODELLED RESPONSE UNDER A SCENARIO · DIFFERENT CLAIM WITH A PREDICTION HORIZON

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
OBSERVATION → measured past or present	SCENARIO → conditional assumptions	FORECAST → different claim with a prediction horizon
ATTRIBUTION → explanation of observed change	PROJECTION → modelled response under a scenario	OBSERVATION → measured past or present

ECOLOGICAL UNIT / PARAMETER
- OBSERVATION → measured past or present
- ATTRIBUTION → explanation of observed change

DECISIVE DISTINCTION
- SCENARIO → conditional assumptions
- PROJECTION → modelled response under a scenario

SCALE / STATUS / EXCEPTION
- FORECAST → different claim with a prediction horizon
- OBSERVATION → measured past or present

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FORECAST → different claim with a prediction horizon.

STAGE 09

OBSERVATION-PROJECTION MATRIX

OBSERVATION-PROJECTION MATRIX · MEASURED PAST OR PRESENT · EXPLANATION OF OBSERVED CHANGE · CONDITIONAL ASSUMPTIONS · MODELLED RESPONSE UNDER A SCENARIO · DIFFERENT CLAIM WITH A PREDICTION HORIZON

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
OBSERVATION → measured past or present	SCENARIO → conditional assumptions	FORECAST → different claim with a prediction horizon
ATTRIBUTION → explanation of observed change	PROJECTION → modelled response under a scenario	OBSERVATION → measured past or present

ECOLOGICAL UNIT / PARAMETER

- OBSERVATION → measured past or present
- ATTRIBUTION → explanation of observed change

DECISIVE DISTINCTION

- SCENARIO → conditional assumptions
- PROJECTION → modelled response under a scenario

SCALE / STATUS / EXCEPTION

- FORECAST → different claim with a prediction horizon
- OBSERVATION → measured past or present

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FORECAST → different claim with a prediction horizon.

STAGE 10

CLIMATE-RESPONSE FORK

CLIMATE-RESPONSE FORK · SOURCE REDUCTION · SINK ENHANCEMENT · MITIGATION WITH ACCOUNTING BOUNDARY · EXPOSURE OR VULNERABILITY REDUCTION · RESIDUAL HARM · SEPARATE LOSS-AND-DAMAGE QUESTION · MITIGATION AND ADAPTATION ARE COMPLEMENTARY

SOURCE REDUCTION → mitigation

SINK ENHANCEMENT → mitigation with accounting boundary → EXPOSURE OR VULNERABILITY REDUCTION → adaptation → RESIDUAL HARM → separate loss-and-damage question

SYSTEM / LEVEL / DEFINITION

- SOURCE REDUCTION → mitigation
- SINK ENHANCEMENT → mitigation with accounting boundary

STRUCTURE-FUNCTION MECHANISM

- EXPOSURE OR VULNERABILITY REDUCTION → adaptation
- RESIDUAL HARM → separate loss-and-damage question

SCALE / EVIDENCE LIMIT

- PORTFOLIO → mitigation and adaptation are complementary
- SOURCE REDUCTION → mitigation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESIDUAL HARM → separate loss-and-damage question; PORTFOLIO → mitigation and adaptation are complementary.

STAGE 11

SCIENCE ANSWER SPINE

SCIENCE ANSWER SPINE · NATURAL EFFECT, HUMAN ENHANCEMENT AND ENERGY FLOW · EMISSIONS, CONCENTRATIONS, STOCK AND NET BALANCE · FORCING, FEEDBACK AND GAS-SPECIFIC PROPERTIES · TREND, DETECTION, ATTRIBUTION AND PROJECTION · SCALE, SCENARIO, CONFIDENCE AND NO INVENTED FIGURES

DEFINE → natural effect, human enhancement and energy flow → ACCOUNT → emissions, concentrations, stock and net balance → EXPLAIN → forcing, feedback and gas-specific properties → EVALUATE → trend, detection, attribution and projection → QUALIFY → scale, scenario, confidence and no invented figures

SYSTEM / LEVEL / DEFINITION

- DEFINE → natural effect, human enhancement and energy flow
- ACCOUNT → emissions, concentrations, stock and net balance

STRUCTURE-FUNCTION MECHANISM

- EXPLAIN → forcing, feedback and gas-specific properties
- EVALUATE → trend, detection, attribution and projection

SCALE / EVIDENCE LIMIT

- QUALIFY → scale, scenario, confidence and no invented figures
- DEFINE → natural effect, human enhancement and energy flow

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUALIFY → scale, scenario, confidence and no invented figures.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · INDIA'S NATIONAL EMISSIONS PROFILE (AS REPORTED IN ITS PERIODIC · ANALYTICAL POINT · INDIA'S PER-CAPITA AND HISTORICAL CUMULATIVE EMISSIONS REMAIN · GLOBAL EMISSIONS AND REMAINING-CARBON-BUDGET FIGURES ARE PERIODICALLY UPDATED BY · ATTRIBUTION-SCIENCE CONFIDENCE LEVELS VARY MEANINGFULLY BY EVENT TYPE (HIGH · INDIA-SPECIFIC SECTORAL EMISSIONS DATA (FROM NATIONAL COMMUNICATIONS/BURS) IS REPORTED · IPCC AR6 ATTRIBUTION SCIENCE HAS SUBSTANTIALLY STRENGTHENED CONFIDENCE IN

OPTIONAL ECOLOGICAL PROCESS DEPTH

- CAUTION: India's national emissions profile (as reported in its periodic UNFCCC national
- CAUTION: Analytical point: India's per-capita and historical cumulative emissions remain
- CAUTION: Global emissions and remaining-carbon-budget figures are periodically updated by IPCC
- CAUTION: Attribution-science confidence levels vary meaningfully by event type (high confidence

SCALE, PARAMETER OR STATUS LIMIT

- CAUTION: India-specific sectoral emissions data (from national communications/BURs) is reported
- IPCC AR6 attribution science has substantially strengthened confidence in linking
- Aerosols exert a net cooling (negative radiative forcing) effect that partially masks
- The remaining carbon budget concept reframes climate policy around a finite, depleting

COMPARATIVE RESTORATION NUANCE

- India's methane and nitrous oxide emissions are disproportionately driven by
- NOT: All extreme weather events can be attributed to climate change with equally high
- NOT: Reducing air pollution (aerosols) has no bearing on the rate of global warming. - It
- NOT: A distant net-zero target date alone is sufficient to remain within a fixed carbon

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels